Motor Torque Optimization Using a Reduced Geometric–Thermal Model

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Abstract

This report studies the geometric optimization of a compact permanent-magnet motor for robotic actuators using a reduced-order analytical model. Five geometric variables are optimized to maximize the thermally limited output torque under practical constraints. The nonlinear optimization problem is solved using IPOPT via CasADi, and the optimality is verified through KKT conditions and feasible direction tests. Results show that the optimal design consistently saturates geometric bounds, favoring large stator size, thick magnets, and a minimal air gap. Sensitivity analysis indicates that thermal and magnetic parameters mainly scale the achievable torque without changing the optimal geometry.

1 Introduction

This project studies the geometric optimization of a compact permanent-magnet motor for robotic actuator applications, with the objective of maximizing the achievable torque at the thermal current limit. The problem is nontrivial because these variables affect electromagnetic and thermal performance in competing ways: increasing size or magnet thickness improves the torque constant but increases copper loss, while reducing the air-gap enhances flux but is limited by mechanical and manufacturing constraints. These competing effects necessitate a nonlinear optimization framework to determine the optimal motor geometry.

2 Problem Statement and Model Formulation

The design variables are the stator outer diameter, stator inner diameter, axial stack length, magnet thickness, and air-gap length. These geometric parameters determine the motor magnetic flux, winding resistance, thermal current limit, and ultimately the maximum output torque. The objective is to identify the motor geometry that yields the maximum torque at the thermal limit while satisfying practical manufacturing, packaging, and mechanical clearance constraints.

2.1 Negative Null Form

minimize	$f(\mathbf{x}) = -K_t(\mathbf{x})\, I_{\max}(\mathbf{x})$	maximize torque capability
with respect to	$\mathbf{x} = \begin{bmatrix} x_D \\ x_I \\ x_L \\ x_T \\ x_G \end{bmatrix}$	x_D stator outer diameter x_I stator inner diameter x_L stack length x_T magnet thickness x_G air–gap
subject to	$g_1(\mathbf{x}) = \ 0.03 - x_D \leq 0$	stator outer diameter lower bound
	$g_2(\mathbf{x}) = \ x_D - 0.04 \leq 0$	stator outer diameter upper bound
	$g_3(\mathbf{x}) = \ 0.007 - x_I \leq 0$	stator inner diameter lower bound
	$g_4(\mathbf{x}) = \ x_I - 0.015 \leq 0$ [1.5pt]	stator inner diameter upper bound
	$g_5(\mathbf{x}) = \ 0.008 - x_L \leq 0$	stack length lower bound
	$g_6(\mathbf{x}) = \ x_L - 0.017 \leq 0$	stack length upper bound
	$g_7(\mathbf{x}) = \ 0.001 - x_T \leq 0$	magnet thickness lower bound
	$g_8(\mathbf{x}) = \ x_T - 0.004 \leq 0$	magnet thickness upper bound
	$g_9(\mathbf{x}) = \ 0.0003 - x_G \leq 0$	air–gap lower bound
	$g_{10}(\mathbf{x}) = \ x_G - 0.0015 \leq 0$	air–gap upper bound
	$g_{11}(\mathbf{x}) = \ 0.002 - (x_D - x_I) \leq 0$	mechanical clearance constraint
	$\mathbf{x} \in \mathbb{R}^5$	decision variables are real number
where	$N_t(\mathbf{x}) = 3\dfrac{x_D - x_I}{2d_w}$	turns per phase
	$N_\phi(\mathbf{x}) = c_N N_t(\mathbf{x})$	effective series turns
	$B_g(\mathbf{x}) = \dfrac{k_B B_r x_T}{x_T/\mu_r + x_G}$	air–gap flux density [2]
	$\tau_p(\mathbf{x}) = \dfrac{\pi x_I}{2p}$	pole pitch
	$\phi_{\text{pole}}(\mathbf{x}) = B_g(\mathbf{x})\tau_p(\mathbf{x})x_L$	flux per pole
	$\psi_m(\mathbf{x}) = k_w N_\phi(\mathbf{x})\phi_{\text{pole}}(\mathbf{x})$	flux linkage
	$K_t = \dfrac{3}{2}\psi_m(\mathbf{x})$	torque constant [1]
	$\ell_{\text{turn}} = \pi\dfrac{x_D + x_I}{2} + (x_D - x_I)$	mean turn length
	$R_\phi(\mathbf{x}) = \dfrac{\rho_{\text{cu}}}{A_w} N_\phi(\mathbf{x})\, \ell_{\text{turn}}$	phase resistance
	$I_{\max}(\mathbf{x}) = \sqrt{\dfrac{2\Delta T_{\max}}{3R_\theta R_\phi(\mathbf{x})}}$	thermal current limit [3][1]

2.2 Constraints(Bounds)

We impose simple bounds on the geometry and one additional inequality to avoid unrealistically small tooth height:

$$\begin{aligned} 0.03 \le x_D \le 0.04, \\ 0.007 \le x_I \le 0.015, \\ 0.008 \le x_L \le 0.017, \\ 0.001 \le x_T \le 0.004, \\ 0.0003 \le x_G \le 0.0015, \\ 0.002 - (x_D - x_I) \le 0. \end{aligned}$$

The last constraint ensures $x_D - x_I \ge 0.002$ m, i.e., sufficient tooth height for winding placement.

2.3 Parameter Table

Fixed parameters are shown in Table 1

Table 1: Model Parameters Used in the Analytical Formulation

Symbol	**Description**	**Base Value**	**Sensitivity Values**	**Units**
p	Pole pairs	7	–	–
k_w	Winding factor	0.92	–	–
k_B	Flux correction factor	0.80	–	–
B_r	Magnet remanence	1.20	{0.8, 1.0, 1.2, 1.4, 1.6}	T
μ_r	Magnet relative permeability	1.05	–	–
ρ_{cu}	Copper resistivity	1.72×10^{-8}	–	Ω·m
R_θ	Thermal resistance	3.0	–	K/W
$\Delta T_{\max}$	Allowable temperature rise	80	{60, 70, 80, 90, 100}	°C
c_N	Phase turn factor	4.0	–	–
d_w	Wire diameter (20 AWG)	0.812	–	mm
A_w	Wire cross-sectional area	0.518	–	mm^2

3 Analysis of Problem Statement

3.1 Assumptions and Validity

To obtain a tractable and fully differentiable optimization model, the following simplifying assumptions are made. The motor pole-pair number is fixed to $p = 7$. The stator tooth height is approximated as $h_{\text{tooth}}(\mathbf{x}) = (x_D - x_I)/2$. A continuous winding model with 20 AWG wire is used, where each tooth is wrapped in three layers, giving the turns per tooth $N_t(\mathbf{x}) = 3(x_D - x_I)/(2d_w)$. The effective phase turns are defined by $N_\phi(\mathbf{x}) = c_N N_t(\mathbf{x})$, where c_N is a constant phase-turn factor. The air-gap flux density is modeled as

$$B_g(\mathbf{x}) = \frac{k_B B_r x_T}{x_T/\mu_r + x_G},$$

where k_B accounts for leakage and fringing and μ_r is the magnet relative permeability. The phase resistance follows the standard copper loss model $R_\phi(\mathbf{x}) = (\rho_{\text{cu}}/A_w)\, N_\phi(\mathbf{x})\, \ell_{\text{turn}}(\mathbf{x})$, and the thermal current limit is given by[3, 4].

$$I_{\max}(\mathbf{x}) = \sqrt{\frac{2\Delta T_{\max}}{3R_\theta R_\phi(\mathbf{x})}}.$$

These assumptions establish a continuous mapping from geometry to winding, magnetic flux, resistance, and thermal current limit, and are valid within the feasible design domain.

3.2 Constraint Assessment

All constraints in this optimization problem are *practical constraints*. The first ten constraints impose lower and upper bounds on the five geometric design variables x_D, x_I, x_L, x_T, and x_G. These bounds reflect engineering packaging limits, manufacturing feasibility, and actuator size requirements rather than physical laws. There are no equality constraints in the formulation; therefore, the problem contains no strict natural constraints arising directly from conservation laws.

3.3 Problem Classification

- **Class:** Smooth nonlinear program (NLP).
- **Continuity:** All functions (objective and constraints) are continuous on the feasible domain.
- **Smoothness:** All functions are differentiable with continuous first derivatives inside the bounds.
- **Convexity:** The problem is nonconvex because the objective includes products and square-root nonlinearity in the decision variables; the Hessian is not globally positive semidefinite.
- **Size:** 5 decision variables, 11 inequality constraints (including simple bounds).

3.4 Scaling Decisions

All variables are implemented directly in consistent SI units. The geometric decision variables are on the order of 10^{-3} to 10^{-2} meters, and the optimal torque is on the order of 10^{-2} N·m, which provides a well-scaled problem for the numerical solver. Therefore, no additional manual scaling or nondimensionalization was applied, and the internal scaling of the `IPOPT` solver was sufficient for robust convergence.

4 Optimization Study

4.1 Numerical Method

We solve the optimization problem using the interior-point solver IPOPT through the CasADi interface in Python. The full motor model—including the geometric relations, magnetic flux expressions,

resistive heating model, and thermal current limit—is implemented symbolically in CasADi, allowing IPOPT to compute exact first- and second-order derivatives automatically. An interior-point method is appropriate for this problem because:

- It handles nonlinear inequality constraints and bound constraints natively.
- It efficiently exploits sparsity in the symbolic derivatives generated by CasADi.
- It provides a robust mechanism for solving smooth, nonconvex NLPs such as this motor design problem.

The optimization is posed in the standard negative-null form

$$\min_{\mathbf{x}} f(\mathbf{x}) = -T(\mathbf{x}),$$

subject to the physical feasibility constraint

$$D_{\text{out}} - D_{\text{in}} \;\geq\; \text{MIN_RADIAL_GAP},$$

together with simple bound constraints on all decision variables. IPOPT then solves this smooth constrained NLP using a barrier-penalty interior-point strategy.

Although IPOPT reports classical KKT quantities (Lagrange multipliers, constraint residuals, stationarity norms), these values do *not* serve as reliable optimality certificates in this problem. At the optimizer, the active radial-gap constraint and several active bound constraints violate the Linear Independence Constraint Qualification (LICQ), making the KKT system ill-posed and causing the multipliers to lose physical or mathematical interpretability. Consequently, KKT conditions cannot be used to verify local optimality.

Instead, optimality is verified using two solver-independent mechanisms: a multi-start global search (all random initializations converge to the same design) and a one-sided feasible directional curvature test showing that torque strictly decreases under all allowable perturbations of the solution.

4.2 Optimal Solution

IPOPT converged to the following optimal geometry:

$$x^\star = \begin{bmatrix} 0.04000 \\ 0.01500 \\ 0.01700 \\ 0.00400 \\ 0.00030 \end{bmatrix} \text{ m}.$$

The corresponding performance at $x^\star$ is:

$$K_t(x^\star) = 0.0068153 \text{ N·m/A}, \qquad I_{\max}(x^\star) = 7.2126 \text{ A},$$
$$T_{\max}(x^\star) = 0.0491562 \text{ N·m}.$$

The gap constraint is satisfied:

$$D_{\text{out}} - D_{\text{in}} - \text{MIN_RADIAL_GAP} = 0.04 - 0.015 - 0.002 = 0.023 \text{ m},$$

so the explicit nonlinear gap constraint is inactive at the optimum.

4.3 Constraint Activity and Lagrange Multipliers

At the optimum: x_D is at its upper bound (4 cm). x_T is near its upper bound. x_G is near its lower bound. This indicates that the optimizer pushes toward larger outer diameter, thicker magnets, and a very small air gap, which matches physical intuition for maximizing flux and torque. The associated Lagrange multipliers (from `trust-constr`) for these bounds are positive, consistent with active constraints. Constraints that are not active at the solution have zero or near-zero multipliers.

4.4 Termination and Local Optimality

IPOPT terminated with the status *"Optimal Solution Found"*. The final summarized residuals reported by IPOPT were:

- Objective value: $f(x^\star) = -4.9156 \times 10^{-2}$,
- Dual infeasibility: 1.8×10^{-12},
- Constraint violation: 0 (within tolerance),
- Variable bound violation: $\approx 1.0 \times 10^{-9}$,
- Complementarity: $\approx 9.1 \times 10^{-11}$.

These values indicate that all first-order KKT conditions (primal feasibility, dual feasibility, and complementarity) are satisfied to near machine precision. In a separate post-processing step, we evaluated the gradient of the Lagrangian. When we ignore bound multipliers and only include the nonlinear constraint multiplier, the Lagrangian gradient norm is large (about 12.5), which is expected because several bound constraints are active. When the bound multipliers from IPOPT are included, the full KKT stationarity condition is satisfied within numerical precision, consistent with the solver's report.

Combined with the positive-definiteness of the reduced Hessian in the active constraint subspace (enforced internally by IPOPT's line search and Hessian regularization), this provides strong evidence that $\mathbf{x}^\star$ is a local minimizer of $f(\mathbf{x})$ (equivalently, a local maximizer of $T(\mathbf{x})$).

Numerical One–Sided Feasible Direction Test. To rigorously verify local optimality despite the violated constraint qualification (the radial-gap inequality is active and several variables lie exactly on simple bounds), we perform a one–sided feasible directional test. For each coordinate direction that moves strictly into the interior of the feasible region (i.e., decreasing x_D, x_I, x_L, x_T, or increasing x_G), we evaluate

$$T(\mathbf{x}^\star + \varepsilon d) \quad \text{for} \quad \varepsilon = 10^{-4},$$

and compare it to $T(x^\star)$. A strict local maximizer must satisfy

$$T(\mathbf{x}^\star + \varepsilon d) < T(\mathbf{x}^\star) \quad \text{for all feasible directions } d.$$

The numerical results obtained from our implementation are shown in Table 2. In every feasible direction, the perturbed torque value is strictly smaller than the optimum value $T(\mathbf{x}^\star) = 0.0491562\,\mathrm{N}\cdot\mathrm{m}$, confirming that $\mathbf{x}^\star$ is a strict local maximizer.

Direction	$T(\mathbf{x}^\star + \varepsilon d)$	Difference $T - T^\star$
Decrease x_D	0.04911451	-4.17×10^{-5}
Decrease x_I	0.04893860	-2.18×10^{-4}
Decrease x_L	0.04886706	-2.89×10^{-4}
Decrease x_T	0.04906437	-9.18×10^{-5}
Increase x_G	0.04798848	-1.17×10^{-3}

Table 2: One–sided feasible direction test. All perturbations strictly decrease the torque, providing practical verification of second–order sufficiency and confirming that $\mathbf{x}^\star$ is a strict local maximizer.

This empirical curvature test replaces the classical second–order sufficient conditions, which cannot be applied here due to the failure of LICQ at the solution. Together with the multi–start experiments (12 random initializations all converging to the same point), these results give strong evidence that the optimal geometry $\mathbf{x}^\star$ represents a true isolated local maximizer of the motor torque objective.

4.5 Global Optimality and Interpretation

Because the problem is nonconvex, we cannot prove global optimality analytically. However:

- Multiple starting points within the feasible domain converged to the same solution.
- The objective trends monotonically with respect to several parameters: larger magnetic cross-section, thinner air gap, and more winding volume all increase torque until bounded by constraints.

These observations suggest that the solution found is likely the global maximizer of thermally limited torque within the prescribed bounds.

Physically, the optimal design uses the largest possible stator diameter, nearly maximal magnet thickness, a very small air gap, and a relatively long axial stack. This matches classical electromagnetic design intuition for a compact, high-torque-density BLDC motor [2, 5].

Evaluate solution

The optimal design lies at the extreme corner of the feasible domain:

$$x_D = 0.040, \quad x_I = 0.015, \quad x_L = 0.017, \quad x_T = 0.004, \quad x_G = 0.0003.$$

Given the simplified torque model, this result is fully consistent with the underlying physics encoded in the optimization problem.

Modeling assumptions. The electromagnetic model assumes:

- linear magnetic material behavior (no saturation),
- constant thermal resistance independent of geometry,

- no limit on copper fill fraction beyond geometric bounds,
- no mechanical constraints such as weight, manufacturability, or demagnetization.

Under these assumptions, torque increases monotonically with increased flux linkage and copper volume and decreases with air gap. Since the model does not penalize large dimensions or thick magnets, the optimizer naturally saturates all variables at their extremes. This confirms that the model behaves exactly as expected: it encourages larger geometry, thicker magnets, and a smaller gap, because no competing physical phenomena are present in the model to push the optimum into the interior.

Physical intuition. The resulting design—a motor that is as large as possible with the smallest allowed air gap—is unsurprising for this simplified formulation. Higher torque is obtained by:

- increasing magnet thickness to raise flux density,
- increasing axial length and machine radius to enlarge coil area,
- maximizing slot height to increase copper content,
- minimizing air gap to reduce magnetic reluctance.

All of these trends are consistent with classical scaling laws for torque production in permanent-magnet motors.

Model validity. Although the solution is intuitive for the idealized model, it would not necessarily be physically optimal for a real motor. In practice, additional effects—e.g. core saturation, magnet cost, structural limits, bearing clearances, thermal dissipation that depends strongly on size, and overall mass constraints—prevent all variables from being pushed to extreme values. Thus, the present result is physically interpretable, but also highlights the need for a richer model if the goal is to obtain a realistic engineering design.

Overall, the solution is fully consistent with the modeling assumptions, and its corner–point nature is an expected consequence of using a monotone objective subject only to simple geometric bounds.

5 Sensitivity Analysis

5.1 Lagrange Multipliers

At the optimal solution, multiple practical constraints are active. The solver returned the following bound multipliers:

$$\lambda_x = [\,0.416,\ 2.168,\ 2.892,\ 0.897,\ -11.962\,], \qquad \lambda_g = -3.95 \times 10^{-9}.$$

The nonlinear radial-gap constraint has a multiplier essentially equal to zero, indicating that this constraint is inactive at the optimizer. In contrast, several bound constraints on the decision variables are strongly active. In particular, the positive multipliers on the upper bounds of the stator outer diameter x_D, stack length x_L, and magnet thickness x_T indicate that relaxing these bounds

would directly increase the achievable torque. The large negative multiplier on the lower bound of the air–gap x_G indicates strong sensitivity of the optimal torque to manufacturing tolerances on the gap.

Because the objective function has units of torque (N·m) and all constraints are imposed on geometric variables with units of length (m), the Lagrange multipliers have units of force (N). Their magnitudes indicate that the design is not limited by internal electromagnetic or thermal relationships, but rather by strict geometric and packaging constraints. In particular, the optimizer consistently drives the geometry to the largest allowable volume and the smallest allowable air–gap, confirming that practical constraints dominate the final performance.

5.2 Parametric Study

A parametric study was conducted to assess the robustness of the optimal solution with respect to key fixed thermal and magnetic parameters. Two parameters were varied: the maximum allowable temperature rise $\Delta T_{\max}$ and the magnet remanence B_r.

Effect of $\Delta T_{\max}$. The temperature rise limit was varied over

$$\Delta T_{\max} \in \{60,\ 70,\ 80,\ 90,\ 100\}^\circ\text{C}.$$

The corresponding optimal torques were

$$T_{\max} = \{0.0426,\ 0.0460,\ 0.0492,\ 0.0521,\ 0.0550\}\ \text{N·m}.$$

For all cases, the optimal geometry remained unchanged at the same active bounds:

$$x_D = 0.04, \quad x_I = 0.015, \quad x_L = 0.017, \quad x_T = 0.004, \quad x_G = 0.0003.$$

From the analytical expression, the thermal current limit scales as $\sqrt{\Delta T_{\max}}$ for fixed geometry. The numerical results closely follow this square-root trend, indicating that improvements in allowable temperature rise directly translate into predictable gains in torque without altering the optimal geometry. This demonstrates that the design is primarily volume-limited rather than thermally limited under the chosen bounds.

Effect of Magnet Remanence B_r. The magnet remanence was varied over

$$B_r \in \{0.8,\ 1.0,\ 1.2,\ 1.4,\ 1.6\}\ \text{T}.$$

The corresponding optimal torques were

$$T_{\max} = \{0.0328,\ 0.0410,\ 0.0492,\ 0.0573,\ 0.0655\}\ \text{N·m}.$$

Again, the optimal geometry remained at the same practical bounds in all cases.

Since the air–gap flux density is approximately proportional to B_r for fixed t_m and g, the torque constant $K_t(x)$ and therefore the maximum torque increase nearly linearly with B_r. The numerical results confirm this linear trend.

Robustness of the Optimal Design. Across all parameter variations, the optimal geometric design remained unchanged and consistently saturated the same practical bounds. Only the achievable torque scaled with improvements in thermal or magnetic parameters. This indicates that the solution is robust with respect to moderate uncertainty in material and thermal properties, while being fundamentally limited by geometric packaging and manufacturing constraints.

6 Conclusions

This project formulated and solved a nonlinear geometric optimization problem for a motor using a reduced analytical model. The optimal solution lies at the geometric boundaries, with maximal stator size and magnet thickness and minimal air-gap length, which is consistent with physical intuition. Sensitivity analysis shows that changes in thermal and magnetic parameters only scale the achievable torque while leaving the optimal geometry unchanged. Although the simplified model provides clear physical insight, additional effects such as magnetic saturation and realistic thermal behavior must be included in future work to obtain practically manufacturable designs.

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